

Woojae Kim

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TECHNICAL SKILLS

Programming/Technical: Python, C++, TypeScript, Java, MySQL, HTML/CSS

Frameworks/Libraries: React, Flask, PyTorch, NumPy, ROS2

Hardware/Systems: ESP32, Arduino, Raspberry Pi, Soldering, Circuit Design

Platforms/Tools: Windows, Linux, Docker, Git/GitHub, VS Code

PROJECTS

Autonomous Robot Navigation | C++, ROS2

October 2025 - January 2026

- Developed a full autonomous navigation stack in **C++** on **ROS2** Humble for a differential-drive robot equipped with a camera and LiDAR.
- Integrated and processing multi-sensor data to implement path planning, static obstacle avoidance, and reliable point-to-point navigation.
- Designed and implemented an **A*-based global path planner**, enabling the robot to compute efficient routes in a 2D occupancy grid.

Delirium Detection Handheld | C++, ESP32

October 2025 - November 2025

- Built an **ESP32**-based handheld device that automates CAM delirium screening, improving assessment consistency and enabling screening in low-resource clinical environments.
- Integrated **Gemini API** to dynamically generate standardized CAM questions.
- Implemented secure local storage and a WiFi access-point export server to deliver assessment CSV files to clinicians via a built-in web interface.

Containerized Home Server | Raspberry Pi, Docker

July 2025

- Built a containerized home server on a **Raspberry Pi** by leveraging **Docker** to self-host and manage services.
- Integrated and configured WireGuard and NextCloud for secure remote access and file storage.

Neural Network Color Classifier | PyTorch, NumPy

October 2025

- Implemented and trained a neural network in **PyTorch** to classify colors as warm or cool.
- Processed and fed training data into the model, achieving **99% accuracy** on the validation set.

EXPERIENCE

Waterloo Experience (WE) Accelerate Program

Jan 26 – Apr 10, 2026

Project Team Member, Microsoft Azure & Artificial Intelligence Fundamentals

Waterloo, ON (remote)

- Designed an Azure-powered medical device evaluation solution by utilizing Azure resources to detect early hardware failures, proactively preventing potential product recalls and ensuring reliability.
- Collaborated closely with a four-person team and an Avanade mentor through regular meetings to develop and pitch the final architecture, achieving an “Excellent” (34/40) final presentation rating.
- Conceptualized detailed system architecture diagrams and wireframes to map out Azure resource integration, ensuring the proposed medical device evaluation model was technically viable.

AWARDS & ACHIEVEMENTS

President’s Scholarship of Distinction

University of Waterloo — Awarded for achieving an admission average of 95%+

2025

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Computer Engineering

September 2025 - June 2030 (anticipated)